

Linear Algebra Notes

Prepared for Quantum Computing Training 2026

Purpose

These notes are a linear algebra companion to the lectures in this quantum computing training course.

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1 Vectors over the Complex Numbers

We often write quantum states as linear combinations of basis vectors with complex number coefficients.

Definition 1.1 (Complex vector space). *A complex vector space is a set of vectors that can be added together and multiplied by complex scalars. The basic example is*

$$\mathbb{C}^n = \left\{ \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} : z_j \in \mathbb{C} \right\}.$$

A vector in $\mathbb{C}^2$ has the form

$$v = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \alpha, \beta \in \mathbb{C}.$$

In quantum mechanics, one often writes the same vector as

$$|v\rangle = \alpha |0\rangle + \beta |1\rangle,$$

where

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The symbols $|0\rangle$ and $|1\rangle$ are basis vectors of $\mathbb{C}^2$ written in *Dirac bra-ket notation*. The set $\{|0\rangle, |1\rangle\}$ is the *computational basis* for a qubit. More generally, a basis for a vector space is a list of vectors that lets us write every vector in vector space as a unique linear combination of the basis vectors.

Definition 1.2 (Basis). *A set $\{v_1, \dots, v_n\}$ is a basis for a vector space V if every $v \in V$ can be written uniquely as*

$$v = c_1 v_1 + \dots + c_n v_n.$$

2 Inner Products, Length, and Normalization

Over $\mathbb{R}^n$, the dot product is $x \cdot y = x_1 y_1 + \dots + x_n y_n$. Over $\mathbb{C}^n$, the correct analogue involves complex conjugating the entries of one of the vectors.

Definition 2.1 (Complex inner product). *For $u, v \in \mathbb{C}^n$, where*

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix},$$

the standard complex inner product is

$$\langle u | v \rangle = u^\dagger v = \bar{u}_1 v_1 + \dots + \bar{u}_n v_n.$$

Here $u^\dagger$ means conjugate transpose.

The norm, or length, of v is

$$\|v\| = \sqrt{\langle v|v \rangle}.$$

If $v = (\alpha, \beta)^T$, then

$$\|v\|^2 = |\alpha|^2 + |\beta|^2.$$

Definition 2.2 (Normalized vector). *A vector v is normalized, or a unit vector, if $\|v\| = 1$.*

For a qubit written as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the normalization condition is $|\alpha|^2 + |\beta|^2 = 1$. Geometrically, this simply says that $|\psi\rangle$ lies on the unit sphere in $\mathbb{C}^2$.

Example 2.1 (Checking normalization). *Let*

$$v = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle.$$

Then

$$\|v\|^2 = \left| \frac{1}{\sqrt{3}} \right|^2 + \left| \sqrt{\frac{2}{3}} \right|^2 = \frac{1}{3} + \frac{2}{3} = 1.$$

So v is normalized.

Definition 2.3 (Orthogonality). *Two vectors u and v are orthogonal if their inner product is zero:*

$$\langle u|v \rangle = 0.$$

Example 2.2.

$$\langle 0|1 \rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0.$$

The vectors $|0\rangle$ and $|1\rangle$ are orthonormal: they are orthogonal and each has norm 1. Another useful pair is

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

They are also orthonormal, since

$$\langle +|-\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1-1}{2} = 0.$$

3 Linear Maps and Matrices

A gate in quantum computing is an example of a linear map. Before thinking about gates, we should understand linear maps themselves.

Definition 3.1 (Linear map). *A function $T : V \rightarrow W$ between vector spaces is linear if*

$$T(av + bw) = aT(v) + bT(w)$$

for all vectors v, w and scalars a, b .

A matrix represents a linear map once bases have been chosen. For example,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

acts on $v = (\alpha, \beta)^T$ by

$$Av = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{bmatrix}.$$

The first column of a matrix is what $|0\rangle$ maps to; the second column is what $|1\rangle$ maps to. Indeed,

$$A|0\rangle = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}, \quad A|1\rangle = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}.$$

This is one of the most important habits in linear algebra: understand a linear map by asking what it does to each basis vector.

Example 3.1 (The bit-flip matrix). *Consider*

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle.$$

So X swaps the two standard basis vectors. (But it is not the same as the SWAP gate, which acts on 2 qubits, not 1!)

Example 3.2 (The Hadamard matrix). *The matrix*

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

has columns

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = |+\rangle, \quad H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = |-\rangle.$$

Thus, as a linear map, H maps the $\{|0\rangle, |1\rangle\}$ basis to the $\{|+\rangle, |-\rangle\}$ basis.

4 Unitary Matrices

Quantum gates are not arbitrary linear maps. They must preserve the length of state vectors. In linear algebra, the matrices that preserve complex inner products are called unitary.

Definition 4.1 (Unitary matrix). *A square complex matrix U is unitary if*

$$U^\dagger U = I.$$

Equivalently, $U^{-1} = U^\dagger$. Here, $U^\dagger = (U^T)^$.*

Proposition 4.1. *If U is unitary, then it preserves inner products:*

$$\langle Uv | Uw \rangle = \langle v | w \rangle.$$

In particular, it preserves norms:

$$\|Uv\| = \|v\|.$$

Proof. Using matrix notation,

$$\langle Uv|Uw\rangle = (Uv)^\dagger(Uw) = v^\dagger U^\dagger U w = v^\dagger I w = v^\dagger w = \langle v|w\rangle.$$

□

Thus, normalized quantum states stay normalized after they are acted upon by unitary gates.

Example 4.1 (Hadamard is unitary). *Since H is real and symmetric, $H^\dagger = H$. Then*

$$H^\dagger H = H^2 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I.$$

5 Eigenvectors and Eigenvalues

Eigenvectors are vectors whose direction is not changed by a linear map.

Definition 5.1 (Eigenvector and eigenvalue). *A nonzero vector v is an eigenvector of A with eigenvalue λ if*

$$Av = \lambda v.$$

Example 5.1 (Eigenvectors of X). *The matrix X has*

$$X|+\rangle = |+\rangle, \quad X|-\rangle = -|-\rangle.$$

Therefore $|+\rangle$ is an eigenvector with eigenvalue 1, and $|-\rangle$ is an eigenvector with eigenvalue -1 .

Eigenvectors matter because they reveal the directions in which a linear transformation is simplest (aka diagonal). For a matrix like X , the standard basis $\{|0\rangle, |1\rangle\}$ makes the action look like a swap, while the $\{|+\rangle, |-\rangle\}$ basis makes the action look diagonal:

$$X|+\rangle = +|+\rangle, \quad X|-\rangle = -|-\rangle.$$

6 Tensor Products of Vectors

A single vector in $\mathbb{C}^2$ describes one two-level system. To describe two such systems together, the correct vector space is not $\mathbb{C}^2 + \mathbb{C}^2$ but the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2$, which is isomorphic to $\mathbb{C}^4$.

Definition 6.1 (Tensor product of two column vectors). *If*

$$a = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}, \quad b = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix},$$

then

$$a \otimes b = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix}.$$

The standard basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ is

$$|0\rangle \otimes |0\rangle, \quad |0\rangle \otimes |1\rangle, \quad |1\rangle \otimes |0\rangle, \quad |1\rangle \otimes |1\rangle.$$

These are abbreviated as

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

With the ordering above,

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Example 6.1.

$$|0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0 \\ 1 \cdot 1 \\ 0 \cdot 0 \\ 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = |01\rangle.$$

Tensor products distribute over addition:

$$(u + v) \otimes w = u \otimes w + v \otimes w,$$

and

$$u \otimes (v + w) = u \otimes v + u \otimes w.$$

They also interact with scalars by

$$(cu) \otimes v = c(u \otimes v) = u \otimes (cv).$$

Example 6.2 (A product state expanded). *Let*

$$|a\rangle = a_0 |0\rangle + a_1 |1\rangle, \quad |b\rangle = b_0 |0\rangle + b_1 |1\rangle.$$

Then

$$\begin{aligned} |a\rangle \otimes |b\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle. \end{aligned}$$

7 Tensor Products of Matrices

Tensor products also combine linear maps. If A acts on one vector space V and B acts on another W , then $A \otimes B$ acts on the combined space $V \otimes W$.

Definition 7.1 (Kronecker product of matrices). *For*

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix},$$

the tensor product is

$$A \otimes B = \begin{bmatrix} aB & bB \\ cB & dB \end{bmatrix} = \begin{bmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{bmatrix}.$$

The key compatibility rule is

$$(A \otimes B)(u \otimes v) = (Au) \otimes (Bv).$$

This identity is the reason tensor products are so useful: separate operations on separate systems become one matrix acting on the joint vector.

Example 7.1 (Applying X to the first factor). *The matrix $X \otimes I$ is*

$$X \otimes I = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

It acts as

$$(X \otimes I) |01\rangle = X |0\rangle \otimes I |1\rangle = |1\rangle \otimes |1\rangle = |11\rangle.$$

Example 7.2 (Applying H to the first factor).

$$(H \otimes I) |00\rangle = H |0\rangle \otimes I |0\rangle = |+\rangle \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

This is just matrix multiplication and distributivity of the tensor product.

8 Dimension Growth

The tensor product multiplies dimensions:

$$\dim(V \otimes W) = \dim(V) \dim(W).$$

Therefore,

$$\underbrace{\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2}_{n \text{ factors}} \cong \mathbb{C}^{2^n}.$$

A vector in this space has 2^n coordinates relative to the standard tensor-product basis. For $n = 3$, the basis vectors are

$$|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle.$$

A general vector is

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle.$$

Algebraically, this is just a vector in a large complex vector space. The special structure is that the basis labels are bit strings.

9 Product Vectors and Non-Product Vectors

A vector in $\mathbb{C}^2 \otimes \mathbb{C}^2$ is a product vector if it can be written as $u \otimes v$. Not every vector can be written this way.

Consider a general two-factor vector

$$|\psi\rangle = a_{00} |00\rangle + a_{01} |01\rangle + a_{10} |10\rangle + a_{11} |11\rangle.$$

If this equals

$$(x_0 |0\rangle + x_1 |1\rangle) \otimes (y_0 |0\rangle + y_1 |1\rangle),$$

then the coefficients must satisfy

$$a_{00} = x_0 y_0, \quad a_{01} = x_0 y_1, \quad a_{10} = x_1 y_0, \quad a_{11} = x_1 y_1.$$

A necessary condition is

$$a_{00}a_{11} = a_{01}a_{10}.$$

For two qubits, this condition is also sufficient, up to edge cases where some entries are zero.

Example 9.1 (A non-product vector). *The vector*

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

has coefficients

$$a_{00} = \frac{1}{\sqrt{2}}, \quad a_{01} = 0, \quad a_{10} = 0, \quad a_{11} = \frac{1}{\sqrt{2}}.$$

Then

$$a_{00}a_{11} = \frac{1}{2}, \quad a_{01}a_{10} = 0.$$

Since these are not equal, $|\Phi^+\rangle$ cannot be written as $u \otimes v$.

This example is often called entangled in quantum mechanics. From the linear-algebra perspective, the important fact is simply that the tensor-product space contains vectors that are not simple products of vectors from the factors.

10 Controlled Operations as Linear Maps

A controlled operation is a linear map defined by its action on basis vectors. The controlled-NOT operation, often abbreviated CNOT, acts on the two-factor basis by

$$\begin{aligned} |00\rangle &\mapsto |00\rangle, \\ |01\rangle &\mapsto |01\rangle, \\ |10\rangle &\mapsto |11\rangle, \\ |11\rangle &\mapsto |10\rangle. \end{aligned}$$

Because a linear map is determined by its action on a basis, this completely defines the matrix.

Using the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, the matrix is

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

The columns are the output basis vectors. For instance, the third column is $|11\rangle$ because CNOT sends $|10\rangle$ to $|11\rangle$.

Example 10.1 (Creating an entangled state using linear maps). *Start with $|00\rangle$. Apply $H \otimes I$:*

$$(H \otimes I) |00\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

Now apply CNOT:

$$\text{CNOT} \left(\frac{|00\rangle + |10\rangle}{\sqrt{2}} \right) = \frac{\text{CNOT} |00\rangle + \text{CNOT} |10\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Everything here is just linearity.

11 Change of Basis

The same vector can have different coordinates in different bases. For example, in the computational basis,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

However, $\{|+\rangle, |-\rangle\}$ constitutes a valid basis for the qubit's state space. In this plus/minus basis,

$$|+\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |-\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

One can also write the standard computational basis vectors $|0\rangle$ and $|1\rangle$ in terms of the plus/minus basis:

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}.$$

The Hadamard matrix is the linear map which allows one to change between the computational and plus/minus bases.

12 Projection and Measurement as Linear Algebra

Although quantum measurement has a physical interpretation, its mathematical description involves projecting the state onto the eigenspaces of the measurement basis. For a normalized vector

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

the coordinate α is the inner product $\alpha = \langle 0|\psi\rangle$ and, similarly, $\beta = \langle 1|\psi\rangle$. The squared magnitudes are

$$|\langle 0|\psi\rangle|^2 = |\alpha|^2, \quad |\langle 1|\psi\rangle|^2 = |\beta|^2.$$

The projection matrices onto the two coordinate axes are

$$P_0 = |0\rangle \langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad P_1 = |1\rangle \langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

They satisfy

$$P_0 + P_1 = I, \quad P_0^2 = P_0, \quad P_1^2 = P_1.$$

Applying these projectors to the state gives

$$P_0 |\psi\rangle = |0\rangle \langle 0|\psi\rangle = \alpha |0\rangle, \quad P_1 |\psi\rangle = |1\rangle \langle 1|\psi\rangle = \beta |1\rangle.$$

Thus each projector extracts the component of $|\psi\rangle$ lying along the corresponding basis direction. The squared length of each projected component is

$$\|P_0 |\psi\rangle\|^2 = |\alpha|^2, \quad \|P_1 |\psi\rangle\|^2 = |\beta|^2.$$

In the measurement interpretation, these squared lengths become the probabilities of the two possible outcomes:

$$\Pr(0) = |\alpha|^2, \quad \Pr(1) = |\beta|^2.$$

Since $|\psi\rangle$ is normalized,

$$|\alpha|^2 + |\beta|^2 = 1,$$

so the probabilities sum to one.

After measurement, the state is no longer the original superposition. If the outcome is 0, the state collapses to the normalized projection

$$\frac{P_0 |\psi\rangle}{\sqrt{\langle\psi| P_0 |\psi\rangle}} = \frac{\alpha |0\rangle}{|\alpha|} \sim |0\rangle,$$

where $\sim$ means equal up to an overall phase. Similarly, if the outcome is 1, the post-measurement state is

$$\frac{P_1 |\psi\rangle}{\sqrt{\langle\psi| P_1 |\psi\rangle}} = \frac{\beta |1\rangle}{|\beta|} \sim |1\rangle.$$

So, in the computational basis, measurement both selects one of the basis directions with probability given by the squared projection length and updates the state to the corresponding basis vector.